LAB 8 Server Configuration (DHCP, DNS & Web Server)

Objectives:

- To configure a server in Cisco Packet Tracer.
- To implement DHCP for automatic IP assignment.
- To configure DNS for domain name resolution.
- To configure a Web Server and access it using a browser.

Software Requirement:

- Cisco Packet Tracer (any recent version)
- Windows/Linux PC

Theory:

Server

A server is a powerful computer that provides services, resources, and data to client devices over a network. In this lab, a single server provides DHCP, DNS, and Web services

DHCP (Dynamic Host Configuration Protocol)

DHCP automatically assigns IP addresses and network parameters (subnet mask, gateway, DNS) to client devices.

Advantages

- Reduces manual configuration
- Prevents IP conflicts
- Saves time and effort
- Centralized IP management

DNS (Domain Name System)

DNS translates human-readable domain names (like www.example.com) into IP addresses so that devices can locate servers easily.

Advantages

- Easy website access
- Eliminates need to remember IP addresses
- Faster communication

Web Server

A web server stores and delivers web pages to clients using HTTP/HTTPS protocols. Clients access web pages through browsers.

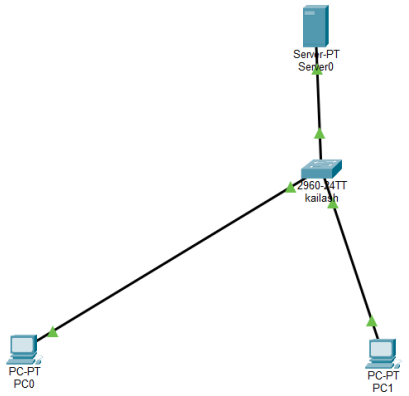
Advantages

- Provides website hosting
- Enables information sharing
- Supports client-server communication

Procedure

1. Create topology in Packet Tracer
2. Connect server and PCs to switch
3. Assign static IP address to server
4. Configure DHCP service on server
5. Configure DNS service on server
6. Configure Web Server (HTTP) service
7. Set PCs to obtain IP automatically
8. Verify DHCP IP assignment on PCs
9. Test DNS using domain name
10. Access web page using browser

Network topology



At the top, there is one server connected to a switch. This server acts as the central device that provides network services such as automatic IP address assignment (DHCP), domain name resolution (DNS), and web hosting (HTTP). In the middle, a switch is used to connect all devices and forward data between them within the local network. At the bottom, two PCs (PC0 and PC1) are connected to the switch and act as client devices that request services from the server.

In this setup, when the PCs start, they obtain IP addresses automatically from the DHCP service running on the server. The DNS service allows the PCs to access the server using a domain name instead of an IP address, and the web server enables the PCs to open and view a web page hosted on the server using a browser. Overall, the topology demonstrates how a single server can provide multiple network services to client devices in a LAN through a switch.

IP Addressing Scheme

Device	IP Address	Subnet Mask	Gateway
Server	192.168.1.2	255.255.255.0	192.168.1.1
PC	192.168.1.1	DHCP	DHCP

PC DHCP Configuration

Device	IPv4 Address	Subnet Mask	Default Gateway	DNS Server

PC0	192.168.1.1	255.255.255.0	0.0.0.0	192.168.1.2
PC1	192.168.1.3	255.255.255.0	0.0.0.0	192.168.1.2

Server Network Settings

Device	Default Gateway	DNS Server
Server0	192.168.1.1	192.168.1.2

1.Server Static IP Configuration:

Steps: Assign a static IP to the server

1. Click Server
2. Go to Desktop → IP Configuration
3. Configure:

IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 192.168.1.2

2. DHCP Configuration:

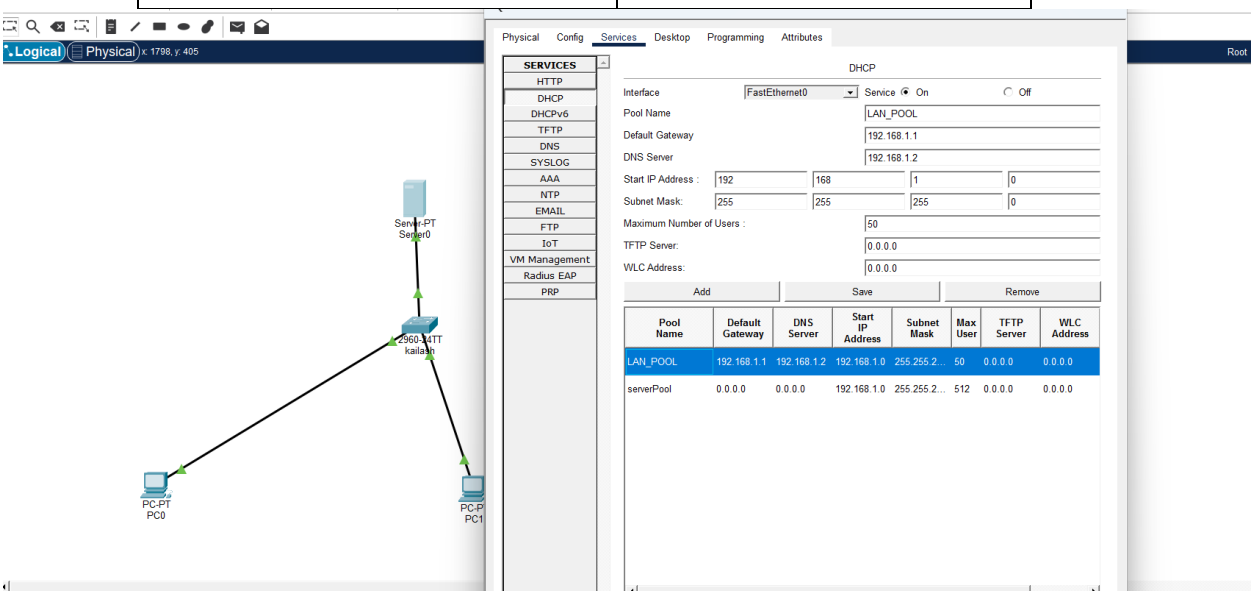
Step 2: Enable DHCP Service

1. Go to Server → Services → DHCP
2. Turn DHCP : ON

Step 3: Create DHCP Pool

Field	Value

Pool Name	LAN_POOL
Default Gateway	192.168.1.1
DNS Server	192.168.1.2
Start IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Maximum Users	50

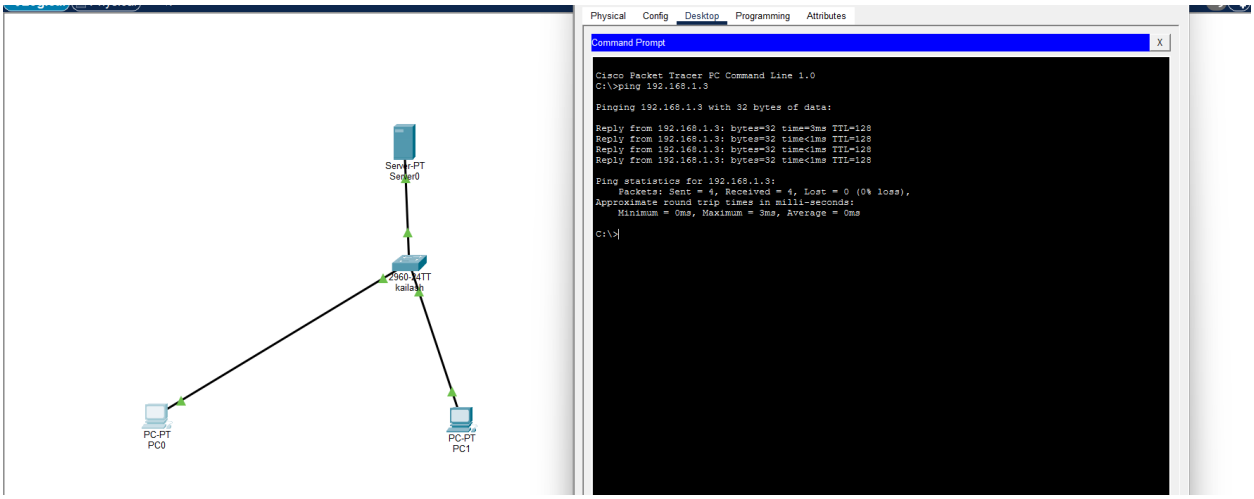


Click **Add** then ,

<u>Verification</u> :

- On each PC: • Go to Desktop → IP Configuration • Select DHCP
- Then test using ping

From pc0 to pc1



3.DNS Server Configuration

Steps4: Enable DNS Service

1. Go to Server → Services → DNS

2. Turn DNS : ON

Step 5: Add DNS Record

Name: www.hcoe.com

Type: A record

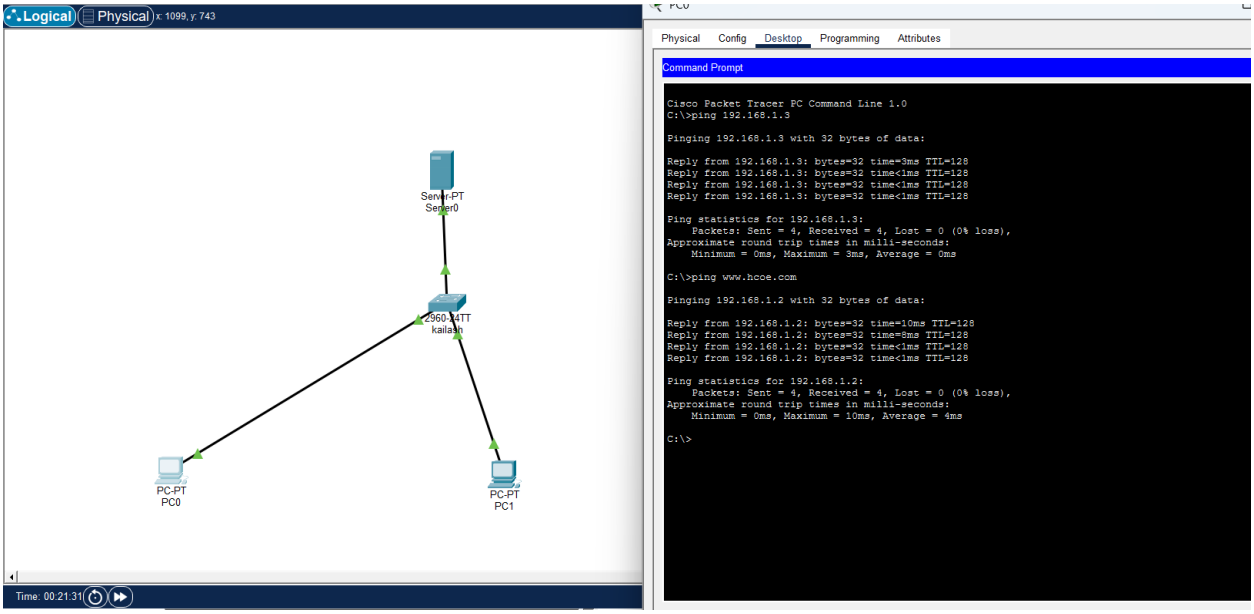
Address: 192.168.1.2

Click ADD

DNS Verification

On each PC:

- Open Command Prompt
- ping www.hcoe.com



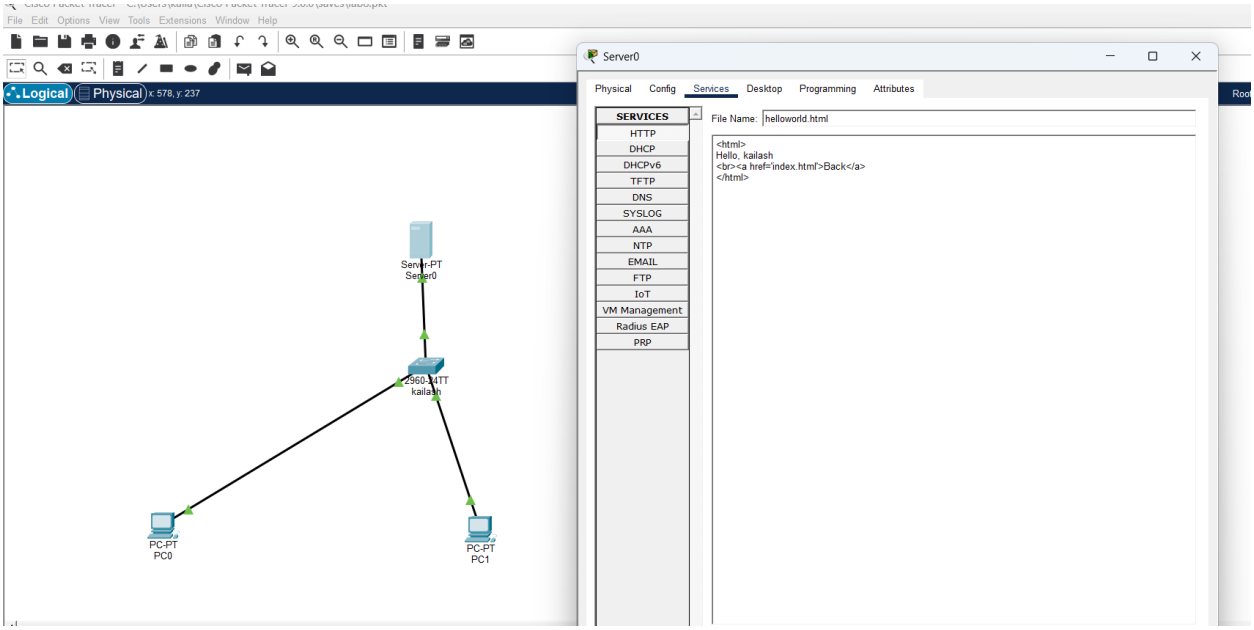
4.Web Server Configuration

Step6: Enable HTTP Service

1. Go to Server → Services → HTTP

2. Turn HTTP : ON

Edit index.html(Optional)

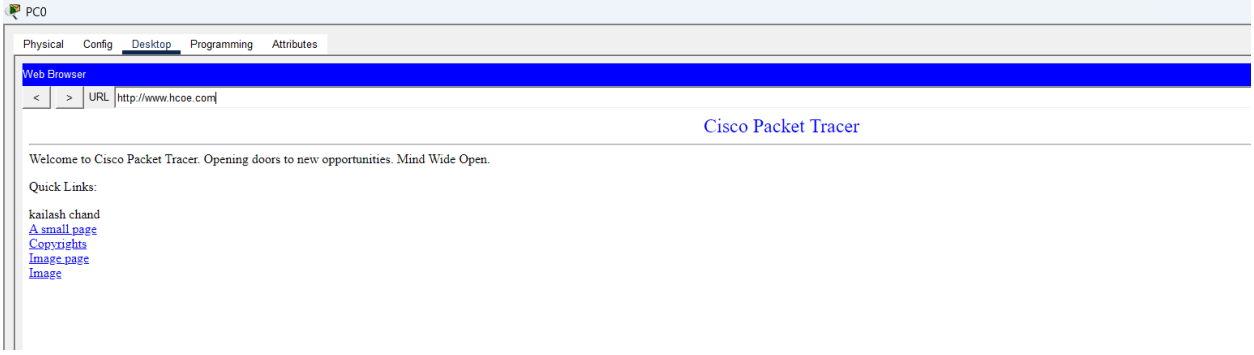


Verification

On PC:

• Go to Desktop → Browser

• Type: http://www.hcoe.com



Result

The server was successfully configured to provide DHCP, DNS, and Web services. PCs obtained IP addresses automatically through DHCP. Domain name resolution worked correctly, and the web page hosted on the server was successfully accessed from client PCs.

Discussion

This lab demonstrated the configuration of essential network services using a single server. DHCP simplified network management by automatically assigning IP addresses, while DNS enabled easy domain name resolution. The web server allowed hosting and accessing web pages within the network. The experiment showed how these services work together in a client-server environment to provide efficient and user-friendly network communication.

Conclusion

The experiment successfully configured DHCP, DNS, and Web services on a server using Cisco Packet Tracer. The lab highlighted the importance of automated IP management, name resolution, and web hosting in modern networks. These services are fundamental for real-world network infrastructure.